A MINIMAL NUMERICAL TYPE WITH MORE THAN ONE VERTEX HAS GENUS AT LEAST THE TOPOLOGICAL GENUS

THE STACKS PROJECT AUTHORS

Editorial note. The counted unit is the Stacks Project Lemma lemma-genus-nonnegative at source lines 911-915 together with its proof at source lines 917-1060, a straight adjacent pairing in the chapter file and the longest proof in this chapter. The chapter is the upstream file models.tex; the pool directory carries the longer name stacks-models and is not used as the chapter name anywhere in this card.

Editorial note. Two context spans carry the definitions the counted statement uses: the lemma that introduces the topological genus at lines 861-866 and the definition of a minimal numerical type at lines 887-893. Both are complete environments from their opening to their closing marker, both are context rather than counted units, and neither overlaps the counted interval.

Editorial note. The retained proof names one lemma of the same chapter at source lines 988 and 1041, the description of minimal types that are not irreducible; it is declared as a non-load-bearing navigation pointer with its stable Stacks tag. The lemma defining the topological genus is retained as context and therefore must NOT also be declared as a pointer: doing so duplicated the label and the engine refused the file with a multiply-defined-label warning.

Editorial note. The author opens the proof with a sentence telling the reader who only cares about types coming from proper regular models that the proof may be skipped, because the geometric case is reproved later in the chapter. That sentence is part of the author's own argument and is retained verbatim; it is not an omission.

Editorial note. The proof is the author's complete argument and contains no omission wording anywhere in its 141 lines.

Editorial note. No citation occurs in the retained spans, so no bibliography entry is supplied.

Allowed standard prerequisites (editorial).

- Numerical types, the topological genus and minimality. A numerical type is a collection of integers n, m of subscript i, a of subscript i j, w of subscript i and g of subscript i satisfying the symmetry and recurrence conditions of the chapter; its topological genus is the expression one minus the number of vertices plus the number of pairs of distinct vertices whose intersection number is positive, and it is nonnegative; the type is minimal when no index has genus zero and self-intersection equal to the negative of its weight, that is, when there is no minus-one index. The author's own definitions are the allowed inputs.. The counted lemma is the lower bound for the genus of a minimal type that the section uses when it classifies types of low genus and when it compares minimal models with their numerical types.

Source reference. Original lemma 0C7B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15326–15326. Navigation pointer invoked by the retained proof at source lines 988 and 1041, where the author recalls the description

Date: stacks-project:models.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

of a minimal type that is not irreducible; the cited statement is an allowed input of the same section and its proof is not reproduced inside the span. The number is the stable Stacks tag of the target, looked up in the shared tag index.

Lemma 0.1. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type. Let e be the number of pairs (i, j) with $i < j$ and $a_{ij} > 0$. Then the expression $g_{top} = 1 - n + e$ is ≥ 0.*

Definition 0.2. We say the numerical type n, m_i, a_{ij}, w_i, g_i of genus g is *minimal* if there does not exist an i with $g_i = 0$ and $a_{ii} = -w_i$, in other words, if there does not exist a (-1)-index.

Lemma 0.3. *If n, m_i, a_{ij}, w_i, g_i is a minimal numerical type with $n > 1$, then $g \geq g_{top}$.*

Proof. The reader who is only interested in the case of numerical types associated to proper regular models can skip this proof as we will reprove this in the geometric situation later. We can write

$$g_{top} = 1 - n + \frac{1}{2}\sum\nolimits_{a_{ij}>0} 1 = 1 + \sum\nolimits_i (-1 + \frac{1}{2}\sum\nolimits_{j\neq i,\ a_{ij}>0} 1)$$

On the other hand, we have

$$\begin{aligned} g &= 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) \\ &= 1 + \sum m_i w_i g_i - \sum m_i w_i + \frac{1}{2}\sum\nolimits_{i\neq j} a_{ij}m_j \\ &= 1 + \sum\nolimits_i m_i w_i(-1 + g_i + \frac{1}{2}\sum\nolimits_{j\neq i} \frac{a_{ij}}{w_i}) \end{aligned}$$

The first equality is the definition, the second equality uses that $\sum a_{ij}m_j = 0$, and the last equality uses that uses $a_{ij} = a_{ji}$ and switching order of summation. Comparing with the formula for g_{top} we conclude that the lemma holds if

$$\Psi_i = m_i w_i(-1 + g_i + \frac{1}{2}\sum\nolimits_{j\neq i} \frac{a_{ij}}{w_i}) - (-1 + \frac{1}{2}\sum\nolimits_{j\neq i,\ a_{ij}>0} 1)$$

is ≥ 0 for each i. However, this may not be the case. Let us analyze for which indices we can have $\Psi_i < 0$. First, observe that

$$(-1 + g_i + \frac{1}{2}\sum\nolimits_{j\neq i} \frac{a_{ij}}{w_i}) \geq (-1 + \frac{1}{2}\sum\nolimits_{j\neq i,\ a_{ij}>0} 1)$$

because a_{ij}/w_i is a nonnegative integer. Since $m_i w_i$ is a positive integer we conclude that $\Psi_i \geq 0$ as soon as either $m_i w_i = 1$ or the left hand side of the inequality is ≥ 0 which happens if $g_i > 0$, or $a_{ij} > 0$ for at least two indices j, or if there is a j with $a_{ij} > w_i$. Thus

$$P = \{i : \Psi_i < 0\}$$

is the set of indices i such that $m_i w_i > 1$, $g_i = 0$, $a_{ij} > 0$ for a unique j, and $a_{ij} = w_i$ for this j. Moreover

$$i \in P \Rightarrow \Psi_i = \frac{1}{2}(-m_i w_i + 1)$$

The strategy of proof is to show that given $i \in P$ we can borrow a bit from Ψ_j where j is the neighbour of i, i.e., $a_{ij} > 0$. However, this won't quite work because j may be an index with $\Psi_j = 0$.

Consider the set

$$Z = \{j : g_j = 0 \text{ and } j \text{ has exactly two neighbours } i, k \text{ with } a_{ij} = w_j = a_{jk}\}$$

For $j \in Z$ we have $\Psi_j = 0$. We will consider sequences $M = (i, j_1, \ldots, j_s)$ where $s \geq 0$, $i \in P$, $j_1, \ldots, j_s \in Z$, and $a_{ij_1} > 0, a_{j_1 j_2} > 0, \ldots, a_{j_{s-1} j_s} > 0$. If our

numerical type consists of two indices which are in P or more generally if our numerical type consists of two indices which are in P and all other indices in Z, then $g_{top} = 0$ and we win by Lemma 0C7B. We may and do discard these cases.

Let $M = (i, j_1, \ldots, j_s)$ be a maximal sequence and let k be the second neighbour of j_s. (If $s = 0$, then k is the unique neighbour of i.) By maximality $k \notin Z$ and by what we just said $k \notin P$. Observe that $w_i = a_{ij_1} = w_{j_1} = a_{j_1j_2} = \ldots = w_{j_s} = a_{j_sk}$. Looking at the definition of a numerical type we see that

$$\begin{aligned} m_i a_{ii} + m_{j_1} w_i &= 0, \\ m_i w_i + m_{j_1} a_{j_1 j_1} + m_{j_2} w_i &= 0, \\ &\ldots\ldots \\ m_{j_{s-1}} w_i + m_{j_s} a_{j_s j_s} + m_k w_i &= 0 \end{aligned}$$

The first equality implies $m_{j_1} \geq 2m_i$ because the numerical type is minimal. Then the second equality implies $m_{j_2} \geq 3m_i$, and so on. In any case, we conclude that $m_k \geq 2m_i$ (including when $s = 0$).

Let k be an index such that we have a $t > 0$ and pairwise distinct maximal sequences $M_1, \ldots, M_t$ as above, with $M_b = (i_b, j_{b,1}, \ldots, j_{b,s_b})$ such that k is a neighbour of j_{b,s_b} for $b = 1, \ldots, t$. We will show that $\Phi_j + \sum_{b=1,\ldots,t} \Phi_{i_b} \geq 0$. This will finish the proof of the lemma by what we said above. Let M be the union of the indices occurring in M_b, $b = 1, \ldots, t$. We write

$$\Psi_k = -\sum\nolimits_{b=1,\ldots,t} \Psi_{i_b} + \Psi'_k$$

where

$$\begin{aligned} \Psi'_k = m_k w_k \Bigg(-1 + g_k + \frac{1}{2} \sum\nolimits_{b=1,\ldots t} \Big(\frac{a_{kj_{b,s_b}}}{w_k} - \frac{m_{i_b} w_{i_b}}{m_k w_k} \Big) + \frac{1}{2} \sum\nolimits_{l \neq k,\ l \notin M} \frac{a_{kl}}{w_k} \Bigg) \\ - \Bigg(-1 + \frac{1}{2} \sum\nolimits_{l \neq k,\ l \notin M,\ a_{kl} > 0} 1 \Bigg) \end{aligned}$$

Assume $\Psi'_k < 0$ to get a contradiction. If the set $\{l : l \neq k,\ l \notin M,\ a_{kl} > 0\}$ is empty, then $\{1, \ldots, n\} = M \cup \{k\}$ and $g_{top} = 0$ because $e = n - 1$ in this case and the result holds by Lemma 0C7B. Thus we may assume there is at least one such l which contributes $(1/2)a_{kl}/w_k \geq 1/2$ to the sum inside the first brackets. For each $b = 1, \ldots, t$ we have

$$\frac{a_{kj_{b,s_b}}}{w_k} - \frac{m_{i_b} w_{i_b}}{m_k w_k} = \frac{w_{i_b}}{w_k} \Big(1 - \frac{m_{i_b}}{m_k} \Big)$$

This expression is $\geq \frac{1}{2}$ because $m_k \geq 2m_{i_b}$ by the previous paragraph and is ≥ 1 if $w_k < w_{i_b}$. It follows that $\Psi'_k < 0$ implies $g_k = 0$. If $t \geq 2$ or $t = 1$ and $w_k < w_{i_1}$, then $\Psi'_k \geq 0$ (here we use the existence of an l as shown above) which is a contradiction too. Thus $t = 1$ and $w_k = w_{i_1}$. If there at least two nonzero terms in the sum over l or if there is one such k and $a_{kl} > w_k$, then $\Psi'_k \geq 0$ as well. The final possibility is that $t = 1$ and there is one l with $a_{kl} = w_k$. This is disallowed as this would mean $k \in Z$ contradicting the maximality of M_1. $\square$